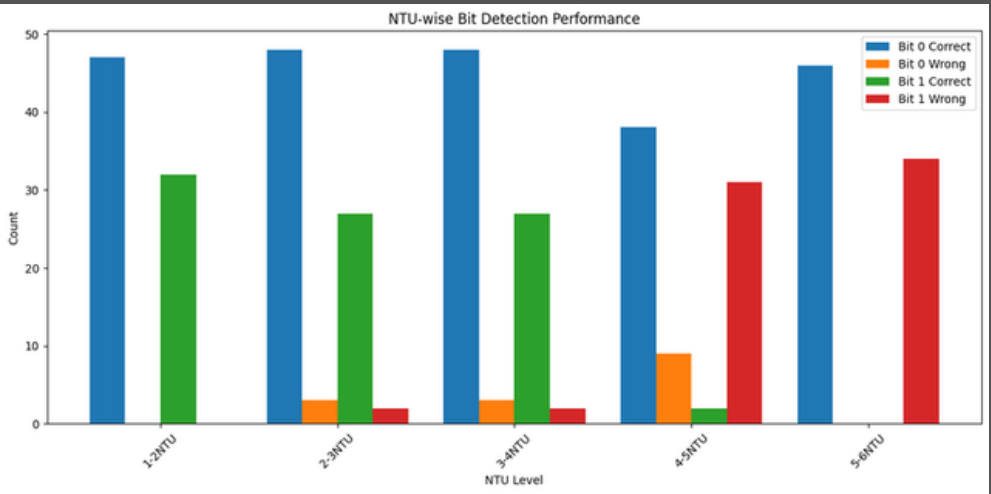
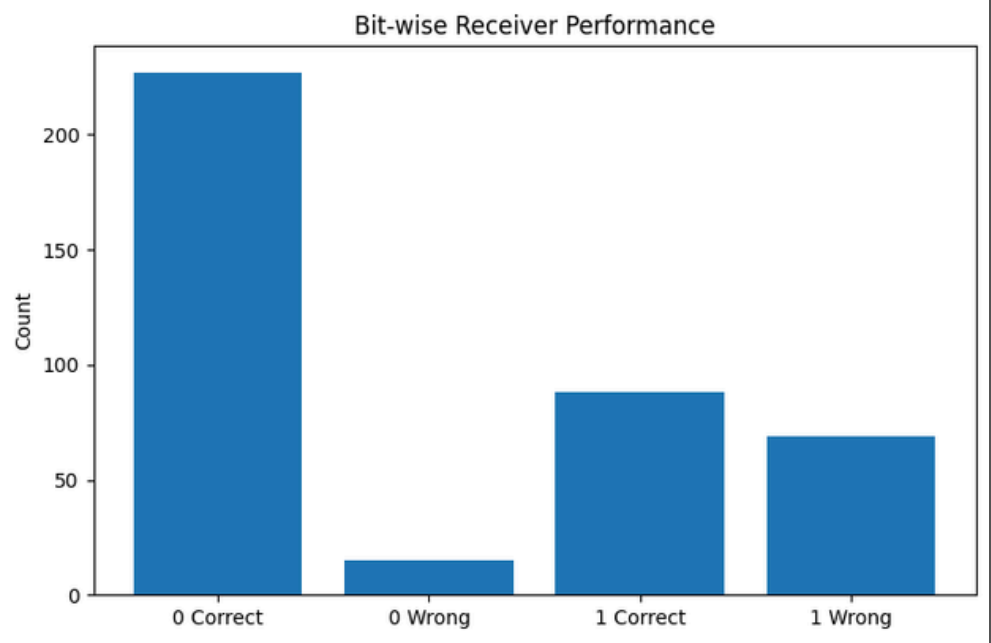
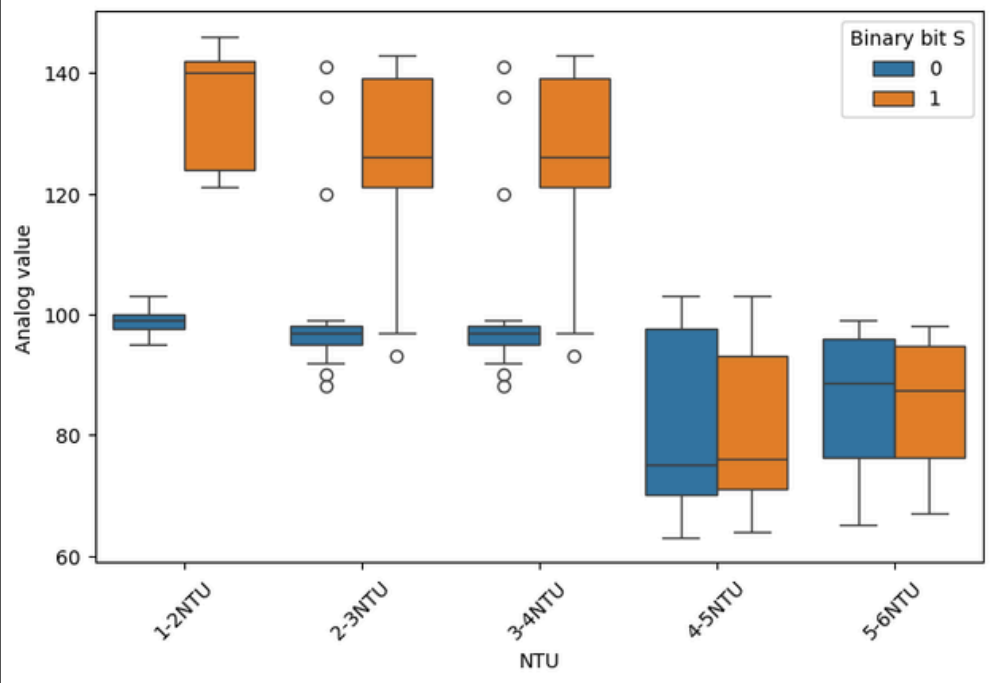
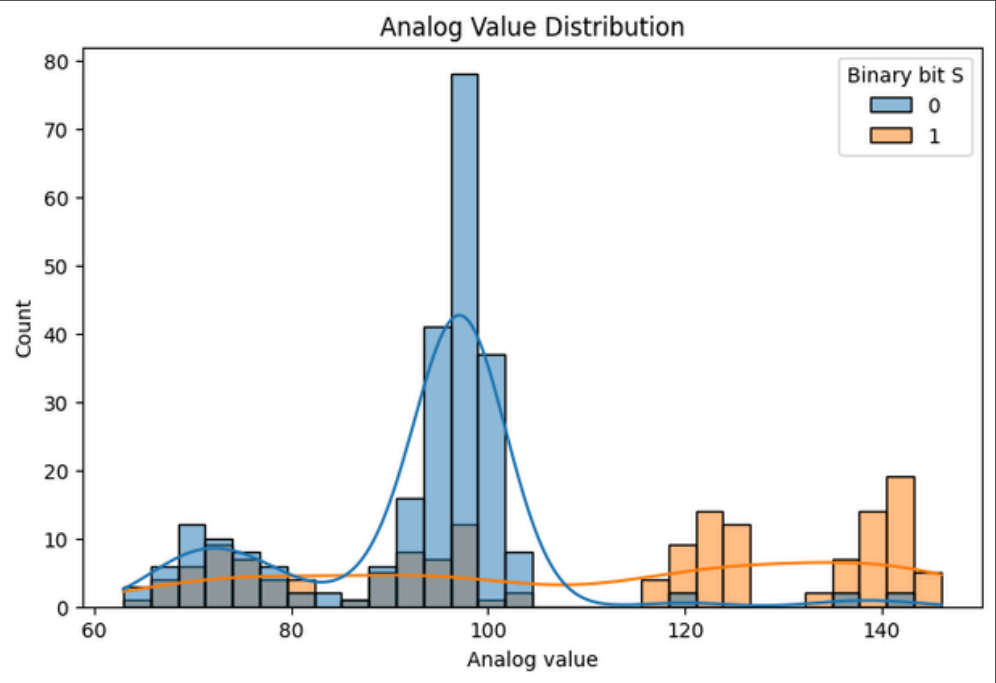
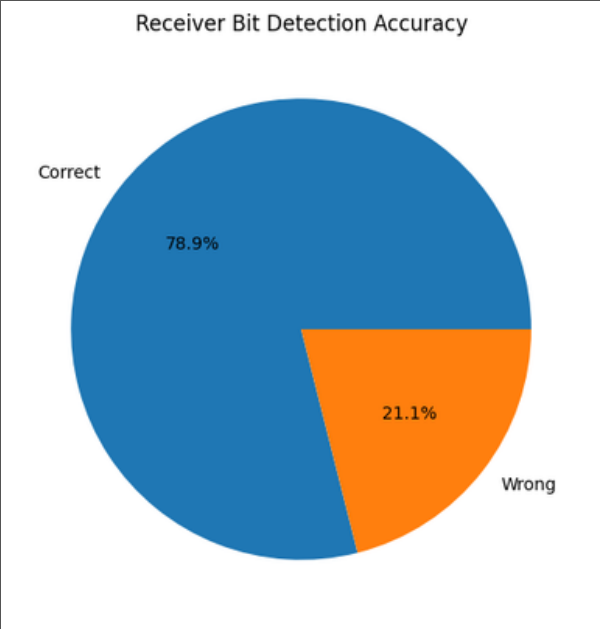
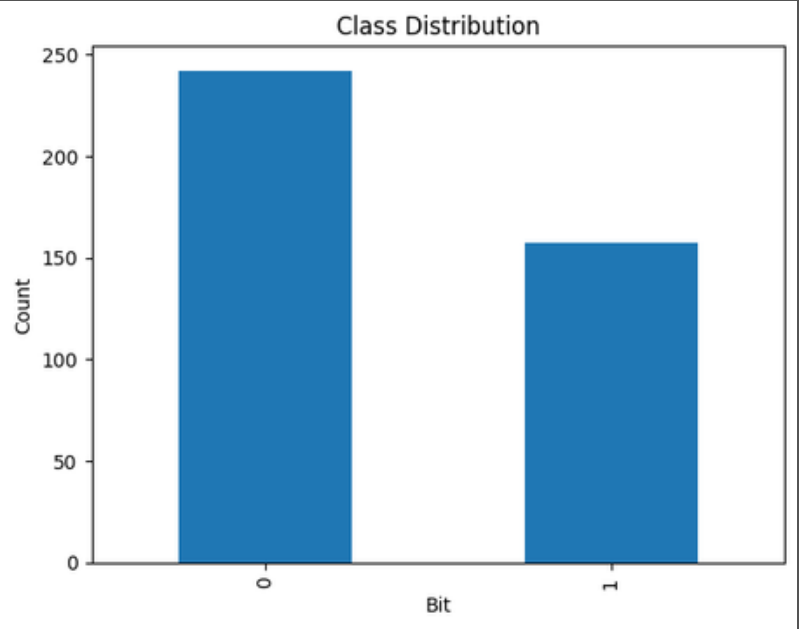


Day 10 - EDA & Class Separability

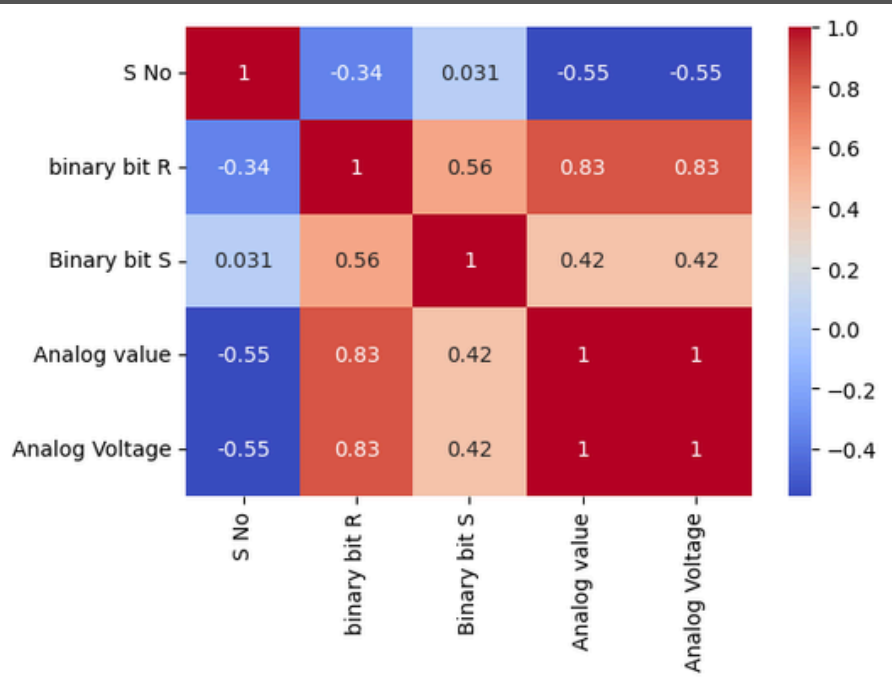


Observation :

- The receiver detects bits correctly 78.9% of the time.
- The test dataset has more Bit 0 samples than Bit 1 samples.
- The system detects Bit 0 much more accurately than Bit 1.
- Detection performance drops significantly as water turbidity increases above 4 NTU.
- High turbidity completely blocks Bit 1 detection, making all signals look like Bit 0.

Observation :

- In clearer water (1-4 NTU), Bit 1 produces much higher analog values than Bit 0, making the two bits easy to separate.
- In murkier water (4-6 NTU), the analog signal drops sharply, causing both bits to overlap and become indistinguishable.
- Overall, Bit 0 stays consistently centered near an analog value of 100, while Bit 1 weakens and scatters across lower values as turbidity rises.



Observation :

Binary Bit s has some moderate correlation ship with the analong voltage and value

